TEACHER GUIDE

Pearson Edexcel GCSE (9-1) in Business (1BS0)

# Break-even

Theme 1: Investigating small business
**Topic 1.3 Putting a business idea into practice · 1.3.2 revenue, costs, profit and break-even**



A revision lesson. Pupils run a car dealership in firms of four and revise break-even on their own deals. Seat each firm together. Each firm answers the topic tasks together.

## The lesson

**45-minute setting**

| Setup | Round 1 | Round 2 | Round 3 | Final questions |
|---|---|---|---|---|

EVERY ROUND

**Firm decisions**
3 min

→

**Deals**
4½–5 min

→

**Topic tasks**
4 min

→

**B2B deal**
2½ min
from Round 2

LESSON SETTINGS

| 30 min | 45 min | 60 min | 75 min |
|---|---|---|---|

# What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



## CEO

**Staff · Sabotage** from Round 2



## Finance

**Capital & Loans · Investments**



## Sales & Marketing

**Marketing**



## Operations

**Seasonal stock**

# Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

## 1 Cars to break even

Round 1

They take up to 8 cars.

**Cars to break even?**

| | |
|---|---|
| Livery and set-up, one-off | **£8,750** |
| Customer pays per car | **£9,750** |
| You pay per car | **£8,000** |

**ANSWER**

**£5**

£9,750 − £8,000 = £1,750;
£8,750 ÷ £1,750 = 5

**Why:** Each car adds £1,750, so 5 cars pay the set-up.

## 2 Lower break-even

Round 1

You want to lower your break-even point.

**Which two changes would lower it?**

- **A** Pay more for each car
- **B** Sell more cars
- **C** Pay higher rent
- **D** Raise the selling price
- **E** Cut the fixed costs

**ANSWER**

**✓ Raise the selling price**
**✓ Cut the fixed costs**

**Why:** More added per car or lower fixed costs; selling more does not move it.

## 3 Rank by break-even

Round 2

Three deals: fixed costs, and what each car adds (price minus cost).

**Order by break-even, fewest cars first?**

- **A** £4,000 fixed, £500 a car
- **B** £9,000 fixed, £1,500 a car
- **C** £6,000 fixed, £1,500 a car

**ANSWER**

**1. £6,000 fixed, £1,500 a car**
**2. £9,000 fixed, £1,500 a car**
**3. £4,000 fixed, £500 a car**

**Why:** Fixed costs ÷ what each car adds: 4, 6 and 8 cars.

# Topic tasks

## 4 Profit on the deal

Round 2

A deal with a set-up cost.

**Profit on the deal?**

| | |
|---|---|
| Livery and set-up, one-off | **£7,000** |
| Customer pays per car | **£10,750** |
| You pay per car | **£9,000** |
| They order | **9 cars** |

**ANSWER**

**£8,750**

9 × (£10,750 − £9,000) − £7,000 = £8,750

**Why:** The first cars pay the set-up; the rest is profit: £8,750.

## 5 Less risky deal

Round 3

Both deals are for 10 cars. The taxi deal breaks even at 4 cars, the hire deal at 8.

**Which deal is less risky, and why?**

**A** The taxi deal

**B** The hire deal

**ANSWER**

**✔ The taxi deal**

**Why:** The taxi deal's margin of safety is 6 cars; the hire deal's only 2.

## 6 New break-even

Round 3

They want £400 off a car.

**New break-even in cars? Round up.**

| | |
|---|---|
| Livery and set-up, one-off | **£5,500** |
| Customer pays per car | **£19,250** |
| You pay per car | **£17,000** |
| Price cut per car | **£400** |

**ANSWER**

**£3**

£18,850 − £17,000 = £1,850; £5,500 ÷ £1,850 → 3

**Why:** Less added per car (price minus cost) means more cars to break even.

# Final questions

Each pupil answers 6 final questions on break-even. They come from these 8 questions.

Then each pupil writes one short answer: “Explain how a higher price changes break-even.”

1 At the break-even level of output, what is true for a dealership?
✔ **Total revenue equals total costs**

2 A dealer’s break-even output is 20 cars a month. In May it sells 24 cars. What does this mean?
✔ **It makes a profit in May**

3 A dealer’s showroom rent rises. Nothing else changes. What happens to its break-even output?
✔ **It rises**

4 A valeting firm raises its price per valet. Nothing else changes. What happens to its break-even output?
✔ **It falls**

5 On a break-even chart, where is the break-even point?
✔ **Where the total revenue line crosses the total costs line**

6 A dealer’s break-even output is 15 cars a month. In March it sells 11 cars. Which statement is correct?
✔ **It makes a loss in March**

7 A cheaper parts supplier cuts the variable cost of each service. Nothing else changes. What happens to break-even output?
✔ **It falls**

8 Why does a new dealer work out its break-even output before opening?
✔ **To know how many cars it must sell to avoid a loss**